

Ordinary differential equations (ODE)

An equation involving one or more ordinary differential coefficients of dependent variable with respect to an independent variable, is called ordinary differential equation. If x is independent variable and y is dependent variable then $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$... denote first order, second order, third order, ... differential coefficients respectively.

Order of the differential equation: The number of highest order derivative is called order of the differential equation.

Degree of the differential equation: The power of highest order derivative is called degree of the differential equation. When the equation has been made free from fraction.

Example:- i. $\frac{dy}{dx} + xy = 0$, 1st order 1st degree
 ii. $x \left(\frac{dy}{dx} \right)^2 + y = 0$, 1st order 2nd degree

iii. $\left\{ \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = 0 \right\}$, 2nd order 1st degree.

iv. $\left(\frac{d^2 y}{dx^2} \right) + ay = 0$, 2nd order 3rd degree.

v. $\sqrt[3]{\left(\frac{d^3 y}{dx^3} \right)^4 - 5x \frac{d^2 y}{dx^2} + y} = \sqrt[5]{\left(\frac{dy}{dx} \right)^2 + y^2 - x}$

$= \left\{ \left(\frac{d^3 y}{dx^3} \right)^4 - 5x \frac{d^2 y}{dx^2} + y \right\}^5 = \left\{ \left(\frac{dy}{dx} \right)^2 + y^2 - x \right\}^3$

3rd order 20 degree.

Linear and non-linear differential equation.

The equation will be called linear differential equation if, it is of first degree in the dependent variable and its partial derivative and product of the dependent variable and its derivatives must be absent. If the equation is not linear, then it is called non-linear differential equation.

Example: i. $\frac{dy}{dx} + xy = x^2 \rightarrow \text{linear}$

ii. $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 2y = 0 \rightarrow \text{linear}$

iii. $x \frac{d^3y}{dx^3} + 2x^2 \frac{d^2y}{dx^2} - y = e^x \rightarrow \text{linear}$

iv. $\frac{dy}{dx^2} + 2 \frac{dy}{dx} + 3y^2 = 0 \rightarrow \text{non-linear}$

v. $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 + y = x \rightarrow \text{non-linear}$

vi. $\frac{d^3y}{dx^3} + y \frac{dy}{dx} + y = e^x \rightarrow \text{non-linear}$

Formation of ODE: $f(x, y, C) = 0$.

1. Show that $x dx + y dy = 0$ is the differential equation of concentric circles whose centres at origin.

Solution: Let, the equation of concentric circles whose centres at origin,

$x^2 + y^2 = a^2 \rightarrow (1)$



Diff. (1) with respect to x ,

$2x + 2y \frac{dy}{dx} = 0 \Rightarrow x + y \frac{dy}{dx} = 0 \Rightarrow x dx + y dy = 0$

Which is the required differential equation. Showed

2. Show that $y dx - x dy = 0$ is the differential equation of any straight lines passing through origin.

Solution: Let, the equation of any straight lines passing through origin,

$$y = mx \rightarrow (1)$$

Differentiating (1) w.r. to x ,

$$\frac{dy}{dx} = m \rightarrow (2)$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} \quad [\text{using (1)}]$$

$$\Rightarrow y dx - x dy = 0.$$

Which is the required differential equation.

3. From the differential equation whose solution $y = A \cos x + B \sin x$ where A and B are arbitrary constants.

Solution: Given that: $y = A \cos x + B \sin x \rightarrow (1)$

Differentiating (1) w.r. to x ,

$$\frac{dy}{dx} = -A \sin x + B \cos x \rightarrow (2)$$

Again, Differentiating (2) w.r. to x ,

$$\therefore \frac{dy}{dx} = -A \cos x - B \sin x$$

$$\Rightarrow \frac{dy}{dx} = -(A \cos x + B \sin x)$$

$$\Rightarrow \frac{dy}{dx} = -y \quad [\text{using (1)}]$$

$\Rightarrow \frac{dy}{dx} + y = 0$, which is the required differential equation.

4. Find the differential equation whose solution is $y^2 = 4a(x+a)$.

Solution: Given that, $y^2 = 4a(x+a) \rightarrow (1)$

Differentiating (1) w.r. to x

$$2y \frac{dy}{dx} = 4a \Rightarrow y \frac{dy}{dx} = 2a \rightarrow (2)$$

$$(1) \Rightarrow y^2 = 4ax + 4a^2$$

$$= 2 \cdot 2a \cdot x + (2a)^2$$

$$\therefore y^2 = 2y \frac{dy}{dx} + \left(y \frac{dy}{dx}\right)^2 \quad [\text{using (2)}]$$

which is the required differential equation.

5. Find the differential equation whose solution $y = e^x (A \cos x + B \sin x)$

6. Find the differential equation whose solution $y = A e^{2x} + B e^{-2x}$ where A and B are arbitrary constants.

7. Show that the differential equation of a family of circles touch the x -axis at origin is $(x^2 - y^2) dy - 2xy dx = 0$

8. Show that the differential equation of a family of circles touch the y -axis at origin is $(x^2 - y^2) dx + 2xy dy = 0$.

9. Show that the D.E of $Ax^2 + By^2 = 1$ is $x \left[y \frac{dy}{dx} + \left(\frac{dy}{dx} \right)^2 \right] = y \frac{dy}{dx}$

5. Solution: Given that, $y = e^x (A \cos x + B \sin x) \rightarrow (1)$

Differentiating (1) w.r. to x , we get

$$\therefore \frac{dy}{dx} = e^x (-A \sin x + B \cos x) + e^x (A \cos x + B \sin x)$$

$$\Rightarrow \frac{dy}{dx} = e^x (-A \sin x + B \cos x) + y \rightarrow (2) \text{ [Using (1)]}$$

Differentiating (2) w.r. to x , we get

$$\therefore \frac{d^2y}{dx^2} = e^x (-A \cos x - B \sin x) + e^x (-A \sin x + B \cos x) + \frac{dy}{dx}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -e^x (A \cos x + B \sin x) + \left(\frac{dy}{dx} - y \right) + \frac{dy}{dx}$$

[Using (2)]

$$\Rightarrow \frac{d^2y}{dx^2} = -y + 2 \frac{dy}{dx} - y \text{ [Using (1)]}$$

$$\Rightarrow \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0.$$

Which is the required differential equation.

6. solution: Given that, $y = Ae^{2x} + Be^{-2x} \rightarrow (1)$

Differentiating (1) w.r. to x , we get

$$\therefore \frac{dy}{dx} = 2Ae^{2x} + 2Be^{-2x} \cdot (-2)$$

$$= 2Ae^{2x} - 2Be^{-2x} \rightarrow (2)$$

Differentiating (2) w.r. to x , we get,

$$\frac{d^2y}{dx^2} = 4Ae^{2x} + 4Be^{-2x}$$

$$\Rightarrow \frac{dy}{dx} = 4y \Rightarrow \frac{dy}{dx} - 4y = 0.$$

Which is the required differential equation.

7. solution: We know, the equation of a family of circles touch the x-axis at origin is -

$$x^2 + y^2 + 2fy = 0 \rightarrow (1)$$

Differentiating (1) w.r to x we get

$$2x + 2y \cdot \frac{dy}{dx} + 2f \frac{dy}{dx} = 0 \rightarrow (2)$$

$$\text{From (1)} \Rightarrow 2f = - \left(\frac{x^2 + y^2}{y} \right) \rightarrow (3)$$

Using (3), from (2)

$$2x + 2y \frac{dy}{dx} - \left(\frac{x^2 + y^2}{y} \right) \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow 2xy + 2y^2 \frac{dy}{dx} - (x^2 + y^2) \frac{dy}{dx} = 0$$

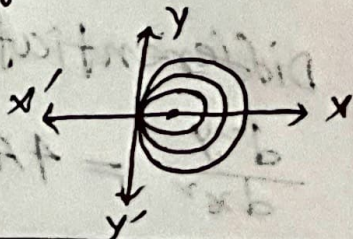
$$\Rightarrow 2xy dx + 2y^2 dy - x^2 dy - y^2 dy = 0$$

$$\Rightarrow (x^2 - y^2) dy - 2xy dx = 0.$$

Which is the required differential equation.

8. solution: We know, the equation of a family of circles touch the y-axis at origin is -

$$x^2 + y^2 + 2gx = 0 \rightarrow (1)$$



Differentiating (1) w.r. to x , we get

$$2x + 2y \frac{dy}{dx} + 2g = 0 \rightarrow (2)$$

$$\text{From (1)} \Rightarrow 2g = -\left(\frac{x^2+y^2}{x}\right) \rightarrow (3)$$

Using (3), from (2) $\Rightarrow$

$$2x + 2y \frac{dy}{dx} - \left(\frac{x^2+y^2}{x}\right) = 0$$

$$\Rightarrow 2x^2 dx + 2y x dy - (x^2+y^2) dx = 0$$

$$\Rightarrow (x^2 - y^2) dx + 2xy dy = 0.$$

Which is the required differential equation.

9. Solution: Given that, $Ax^r + By^r = 1 \rightarrow (1)$

Differentiating (1) w.r. to x , we get

$$2Ax + 2By \frac{dy}{dx} = 0 \Rightarrow Ax + By \frac{dy}{dx} = 0 \rightarrow (2)$$

$$\text{Again, } A + By \frac{d^2y}{dx^2} + B \cdot \left(\frac{dy}{dx}\right)^r = 0 \rightarrow (3)$$

$$\text{From (2)} \Rightarrow \frac{A}{B} = -\frac{y}{x} \frac{dy}{dx} \rightarrow (4)$$

$$\text{From (3)} \Rightarrow \frac{A}{B} = -y \frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^r \rightarrow (5)$$

$$\text{From (4) \& (5), we get } y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^r = \frac{y}{x} \frac{dy}{dx}$$

$$\therefore x \left[y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^r \right] = y \frac{dy}{dx}$$

Which is the required differential equation.

First order and first degree differential equation

variables separable equation: If the differential equation $Mdx + Ndy = 0$ can be expressed to the form $f_1(x) dx = f_2(y) dy$ then it is called variables separable equation.

solve: 1. $\frac{dy}{dx} + 1 = y^2$ 2. $\frac{dy}{dx} + \sqrt{\frac{1-y}{1-x}} = 0$

3. $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$

4. $(e^x + 1)y dy = (y+1)e^x dx$

5. $y\sqrt{x^2-1} dx + x\sqrt{y^2-1} dy = 0$

6. $x\sqrt{1-y^2} dx + y\sqrt{1-x^2} dy = 0$

7. $(x+y) dy + (x-y) dx = 0$

1. solution: $\frac{dy}{dx} + 1 = y^2$

$$\Rightarrow \frac{dy}{dx} = y^2 - 1$$

$$\Rightarrow \frac{dy}{y^2-1} = dx$$

$$\Rightarrow \frac{1}{2} \ln \left| \frac{y-1}{y+1} \right| = x + C$$

which is the required solution:

2. solution: $\frac{dy}{dx} + \sqrt{\frac{1-y}{1-x}} = 0$

$$\Rightarrow \frac{dy}{\sqrt{1-y}} + \frac{dx}{\sqrt{1-x}} = 0$$

$$\Rightarrow \int \frac{dy}{\sqrt{1-y}} + \int \frac{dx}{\sqrt{1-x}} = 0$$

$$\Rightarrow \sin^{-1} y + \sin^{-1} x = C.$$

Which is the required solution.

3. solution: $\sec^x x \tan y \, dx + \sec^y y \tan x \, dy = 0$

$$\Rightarrow \int \frac{\sec^x x}{\tan x} \, dx + \int \frac{\sec^y y}{\tan y} \, dy = 0$$

$$\Rightarrow \ln|\tan x| + \ln|\tan y| = \ln C$$

$$\Rightarrow \tan x \tan y = C.$$

Which is the required solution.

4. solution: $(e^{x+1})y \, dy = (y+1)e^x \, dx$

$$\Rightarrow \frac{y \, dy}{y+1} = \frac{e^x \, dx}{e^{x+1}}$$

$$\Rightarrow \frac{y+1-1}{y+1} \, dy = \frac{e^x}{e^{x+1}} \, dx$$

$$\Rightarrow \left(1 - \frac{1}{y+1}\right) dy = \frac{e^x}{e^{x+1}} \, dx$$

$$\Rightarrow y - \ln(y+1) = \ln(e^{x+1}) + C.$$

Which is the required solution.

5. solution: $y\sqrt{x^2-1} dx + x\sqrt{y^2-1} dy = 0$

$$\Rightarrow \frac{y\sqrt{x^2-1}}{x} dx + \frac{\sqrt{y^2-1}}{y} dy = 0$$

$$\Rightarrow \frac{x^2-1}{x\sqrt{x^2-1}} dx + \frac{y^2-1}{y\sqrt{y^2-1}} dy = 0$$

$$\Rightarrow \frac{x^2}{x\sqrt{x^2-1}} dx + \frac{y^2}{y\sqrt{y^2-1}} dy = \frac{1}{x\sqrt{x^2-1}} dx + \frac{1}{y\sqrt{y^2-1}} dy$$

$$\Rightarrow \frac{1}{2} \cdot \frac{2x}{\sqrt{x^2-1}} dx + \frac{1}{2} \cdot \frac{2y}{\sqrt{y^2-1}} dy = \frac{1}{x\sqrt{x^2-1}} dx + \frac{1}{y\sqrt{y^2-1}} dy$$

$$\Rightarrow \frac{1}{2} \cdot 2\sqrt{x^2-1} + \frac{1}{2} \cdot 2\sqrt{y^2-1} = \sec^{-1}x + \sec^{-1}y + C$$

$$\Rightarrow \sqrt{x^2-1} + \sqrt{y^2-1} = \sec^{-1}x + \sec^{-1}y + C.$$

Which is the required solution.

6. solution: $x\sqrt{1-y^2} dx + y\sqrt{1-x^2} dy = 0$

$$\Rightarrow \frac{x}{\sqrt{1-x^2}} dx + \frac{y}{\sqrt{1-y^2}} dy = 0$$

$$\Rightarrow -\frac{1}{2} \cdot \frac{-2x}{\sqrt{1-x^2}} dx - \frac{1}{2} \cdot \frac{-2y}{\sqrt{1-y^2}} dy = 0$$

$$\Rightarrow -\frac{1}{2} \cdot 2\sqrt{1-x^2} - \frac{1}{2} \cdot 2\sqrt{1-y^2} = -C$$

$$\Rightarrow \sqrt{1-x^2} + \sqrt{1-y^2} = C.$$

Which is the required solution.

7. solution: $(x+y) dy + (x-y) dx = 0$

$$\Rightarrow x dy + y dy + x dx - y dx = 0$$

$$\Rightarrow x dy - y dx + d(xy) = 0$$

$$\Rightarrow \frac{dy}{y} - \frac{dx}{x} + \frac{d(xy)}{xy} = 0$$

$$\Rightarrow \ln y - \ln x + \ln(xy) = \ln c$$

$$\Rightarrow \ln\left(\frac{y \cdot xy}{x}\right) = \ln c$$

$$\Rightarrow y^2 = c$$

Which is the required solution.

Homogeneous Differential Equation

The differential equation $M(x,y)dx + N(x,y)dy = 0$ is called a dimensional homogeneous differential equation if and only if,

$$M(x,y) = x^n m_1\left(\frac{y}{x}\right), \quad N(x,y) = x^n n_1\left(\frac{y}{x}\right)$$

$$\text{or, } M(x,y) = y^n m_1\left(\frac{x}{y}\right), \quad N(x,y) = y^n n_1\left(\frac{x}{y}\right).$$

Q. When $Mdx + Ndy = 0$ is homogeneous. show that the transformation $y = vx$ will separate the variables.

Solution: Let, $Mdx + Ndy = 0$ is the homogeneous differential equation of order n . So, we can write

$$M(x,y)dx + N(x,y)dy = x^n \left[m_1\left(\frac{y}{x}\right)dx + n_1\left(\frac{y}{x}\right)dy \right] = 0$$

$$\Rightarrow m_1\left(\frac{y}{x}\right)dx + n_1\left(\frac{y}{x}\right)dy = 0 \rightarrow (1)$$

$$\text{Let, } y = vx \rightarrow (2)$$

$$\therefore dy = vdx + xdv \rightarrow (3)$$

Using (2) & (3), from (1) $\Rightarrow$

$$m_1(v)dx + n_1(v)(vdx + xdv) = 0$$

$$\Rightarrow \{m_1(v) + v n_1(v)\}dx + x n_1(v)dv = 0$$

$$\Rightarrow \frac{dx}{x} + \frac{n_1(v)}{m_1(v) + v n_1(v)} dv = 0 \rightarrow (4)$$

$\therefore$ Equation (4) is variable separable.

Q: Solve the homogeneous differential equation,

$$1. \frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$$

$$2. (x^r + y^r) dx = 2xy dy$$

$$3. (x+y) dy + (x-y) dx = 0$$

$$4. 2 \frac{dy}{dx} = \frac{y}{x} + \frac{y^r}{x^r}$$

$$5. (x^r + y^r) \frac{dy}{dx} = xy$$

$$6. x dy - y dx = \sqrt{x^2 + y^2} dx$$

$$7. (x^r - y^r) dx + 2xy dy = 0$$

$$8. \frac{dy}{dx} + 2 \frac{y}{x} = \frac{y^3}{x^3}$$

1. Solution: $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x} \rightarrow (1)$

Let, $y = vx$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \rightarrow (2)$$

From (1) and (2), we get

$$v + x \frac{dv}{dx} = v + \tan v$$

$$\Rightarrow x \frac{dv}{dx} = \tan v$$

$$\Rightarrow \frac{dv}{\tan v} = \frac{dx}{x}$$

$$\Rightarrow \cot v dv = \frac{dx}{x}$$

$$\Rightarrow \int \cot v \, dv = \int \frac{dx}{x}$$

$$\Rightarrow \ln \sin v = \ln x + \ln C$$

$$\Rightarrow \sin v = Cx$$

$\therefore \sin \frac{y}{x} = Cx$, Which is the required solution.

2. Solution: $(x^r + y^r) dx = 2xy dy$

$$\Rightarrow \frac{dy}{dx} = \frac{x^r + y^r}{2xy} \rightarrow (1)$$

Let, $y = vx$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \rightarrow (2)$$

From (1) and (2), We get

$$v + x \frac{dv}{dx} = \frac{x^r + v^r x^r}{2x \cdot vx} \quad | \because y = vx$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1 + v^r}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^r}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^r - 2v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 - v^r}{2v}$$

$$\Rightarrow \frac{2v}{v^r - 1} dv + \frac{dx}{x} = 0 \quad [\text{Integrating on both sides}]$$

$$\Rightarrow \ln(v^r - 1) + \ln x = \ln C$$

$$\Rightarrow x \left(\frac{y^2}{x^2} - 1 \right) = C$$

$\Rightarrow y^2 - x^2 = Cx$, which is the required solution.

3. Solution: $(x+y) dy + (x-y) dx = 0$

$$\Rightarrow \frac{dy}{dx} = - \left(\frac{x-y}{x+y} \right) = \frac{y-x}{x+y} \rightarrow (1)$$

Let, $y = vx$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \rightarrow (2)$$

From (1) and (2), we get

$$v + x \frac{dv}{dx} = \frac{vx - x}{x + vx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v-1}{1+v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v-1}{1+v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v-1-v-v^2}{1+v}$$

$$\Rightarrow \frac{v+1}{v^2+1} dv + \frac{dx}{x} = 0$$

$$\Rightarrow \frac{1 \cdot 2v}{2} dv + \frac{1}{v^2+1} dv + \frac{dx}{x} = 0$$

$$\Rightarrow \frac{1}{2} \ln(v^2+1) + \tan^{-1} v + \ln x = \ln C \quad [\text{Integrating}]$$

$$\Rightarrow \frac{1}{2} \ln \left(\frac{y^2}{x^2} + 1 \right) + \tan^{-1} \left(\frac{y}{x} \right) + \ln x = \ln C$$

Which is the required solution.

4. Solution: $2 \frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{x^2}$

$$\Rightarrow \frac{dy}{dx} = \frac{xy + y^2}{2x^2} \longrightarrow (1)$$

Let, $y = vx$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \longrightarrow (2)$$

From (1) and (2), we get:

$$v + x \frac{dv}{dx} = \frac{x \cdot vx + v^2 x^2}{2x^2}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 + v}{2} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - v}{2}$$

$$\Rightarrow \frac{2}{v(v-1)} dv = \frac{dx}{x}$$

$$\Rightarrow 2 \left(\frac{1}{v-1} - \frac{1}{v} \right) dv = \frac{dx}{x}$$

$$\Rightarrow 2 [\ln(v-1) - \ln v] = \ln x + \ln C \quad \text{[Integrating on both sides]}$$

$$\Rightarrow 2 \ln \left(\frac{v-1}{v} \right) = \ln Cx$$

$$\Rightarrow \left(\frac{v-1}{v} \right)^2 = Cx$$

$$\Rightarrow (v-1)^2 = Cx v^2$$

$$\Rightarrow \left(\frac{y}{x} - 1 \right)^2 = Cx \cdot \frac{y^2}{x^2}$$

$$\Rightarrow (y-x)^2 = Cx y^2$$

Which is the required solution.

5. Solution: $(x^2 + y^2) \frac{dy}{dx} = xy \rightarrow (1)$

$\Rightarrow \frac{dy}{dx} = \frac{xy}{x^2 + y^2} \rightarrow (2)$

Let, $y = vx$

$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \rightarrow (3)$

From (2) & (3), we get

$v + x \frac{dv}{dx} = \frac{vx \cdot vx}{x^2 + v^2 x^2} = \frac{v^2}{1 + v^2}$

$\Rightarrow v + x \frac{dv}{dx} = \frac{v^2}{1 + v^2}$

$\Rightarrow x \frac{dv}{dx} = \frac{v^2}{1 + v^2} - v$

$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - v - v^3}{1 + v^2}$

$\Rightarrow \frac{1 + v^2}{v^3} dv + \frac{dx}{x} = 0$

$\Rightarrow \left(\frac{1}{v^3} + \frac{1}{v} \right) dv + \frac{dx}{x} = 0$

$\Rightarrow -\frac{1}{2v^2} + \ln v + \ln x = \ln c$ [Integrating on both sides]

$\Rightarrow -\frac{1}{2} \cdot \frac{x^2}{y^2} + \ln \frac{y}{x} + \ln x = \ln c$

$\Rightarrow -\frac{1}{2} \frac{x^2}{y^2} + \ln y = \ln c$

Which is the required solution.

G. Solution: $x dy - y dx = \sqrt{x^2 + y^2} dx$

$$\Rightarrow x dy = (y + \sqrt{x^2 + y^2}) dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} \quad \text{--- (1)}$$

Let, $y = vx$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \quad \text{--- (2)}$$

From (1) and (2), we get

$$v + x \frac{dv}{dx} = \frac{vx + \sqrt{x^2 + v^2 x^2}}{x}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{(v + \sqrt{1 + v^2})x}{x} - v$$

$$\Rightarrow x \frac{dv}{dx} = v + \sqrt{1 + v^2} - v$$

$$\Rightarrow x \frac{dv}{dx} = \sqrt{1 + v^2}$$

$$\Rightarrow \frac{dv}{\sqrt{1 + v^2}} = \frac{dx}{x}$$

$$\Rightarrow \ln(v + \sqrt{1 + v^2}) = \ln x + \ln C \quad \text{[Integrating on both sides]}$$

$$\Rightarrow v + \sqrt{1 + v^2} = Cx$$

$$\Rightarrow \frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} = Cx$$

$$\Rightarrow y + \sqrt{x^2 + y^2} = Cx^2$$

Which is the required solution.

7. Solution: $(x^2 - y^2) dx + 2xy dy = 0$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy} \rightarrow (1)$$

Let, $y = vx$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \rightarrow (2)$$

From (1) and (2), we get

$$\therefore v + x \frac{dv}{dx} = \frac{v^2 x^2 - x^2}{2x \cdot vx}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1 - 2v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-1 - v^2}{2v}$$

$$\Rightarrow \frac{2v}{1+v^2} dv + \frac{dx}{x} = 0$$

$$\Rightarrow \ln(1+v^2) + \ln x = \ln e \quad [\text{Integrating on both sides}]$$

$$\Rightarrow \left(1 + \frac{y^2}{x^2}\right) \cdot x = e$$

$\Rightarrow x^2 + y^2 = ex$. Which is the required solution.

8. Solution: $\frac{dy}{dx} + 2 \frac{y}{x} = \frac{y^3}{x^3} \rightarrow (1)$

Let, $y = vx$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \rightarrow (2)$$

From (1) and (2), we get

$$\therefore v + x \frac{dv}{dx} = v^3 - 2v$$

$$\Rightarrow x \frac{dv}{dx} = v^3 - 3v$$

$$\Rightarrow \frac{dv}{v(v-3)} = \frac{dx}{x}$$

$$\Rightarrow \frac{dv}{v} // \frac{1}{3} \left(\frac{v}{v-3} - \frac{1}{v} \right) dv = \frac{dx}{x}$$

$$\Rightarrow \frac{1}{3} \left(\frac{1}{2} \cdot \frac{2v}{v-3} - \frac{1}{v} \right) dv = \frac{dx}{x}$$

$$\Rightarrow \frac{1}{6} \ln(v-3) - \frac{1}{3} \ln v = \ln x + \ln C \quad [\text{Integrating on both sides}]$$

$$\Rightarrow \frac{1}{6} \ln \left(\frac{y^2}{x^2} - 3 \right) - \frac{1}{3} \ln \frac{y}{x} = \ln x + \ln C$$

$$\Rightarrow \ln \left(\frac{y^2 - 3x^2}{x^2} \right)^{1/6} = \ln C \cdot x \left(\frac{y}{x} \right)^{1/3}$$

$$\Rightarrow \left(\frac{y^2 - 3x^2}{x^2} \right)^{1/6} = C x \left(\frac{y}{x} \right)^{1/3}$$

$$\Rightarrow y^2 - 3x^2 = C^6 \cdot x^2 \cdot x^6 \cdot \frac{y^2}{x^2}$$

$$\Rightarrow y^2 - 3x^2 = C^6 x^6 y^2$$

Which is the required solution.

Exact Differential Equation

The differential equation $M dx + N dy = 0$ is called exact differential equation if

$M dx + N dy = du$. Where $M = M(x, y)$, $N = N(x, y)$, and $u = u(x, y)$.

Q. Determine the necessary and sufficient condition for $M(x, y) dx + N(x, y) dy = 0$ to be exact differential equation.

Solution: Necessary condition: $M dx + N dy = 0$ is an exact DE if $M(x, y) dx + N(x, y) dy = du(x, y)$

$$\therefore M(x, y) dx + N(x, y) dy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$\Rightarrow \frac{\partial u}{\partial x} = M \rightarrow (1) \quad , \quad \frac{\partial u}{\partial y} = N \rightarrow (2)$$

$$\therefore \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial M}{\partial y} \rightarrow (3) \quad \text{and} \quad \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial N}{\partial x} \rightarrow (4)$$

$$\text{From (3) \& (4)} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Sufficient condition: Now, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow (5)$

$$\text{Let, } \phi = \int M dx + F(y)$$

$$\Rightarrow \frac{\partial \phi}{\partial x} = M \Rightarrow \frac{\partial^2 \phi}{\partial y \partial x} = \frac{\partial M}{\partial y} \rightarrow (5)$$

$$\therefore \frac{\partial N}{\partial x} = \frac{\partial^2 \phi}{\partial x \partial y} \Rightarrow N = \frac{\partial \phi}{\partial y} + V'(y)$$

$$\begin{aligned}\therefore M dx + N dy &= \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \psi'(y) dy \\ &= d\phi + \psi'(y) dy \\ &= d[\phi + \psi(y)] \text{ is exact.}\end{aligned}$$

Q. Determine the general function $N(x, y)$ such that the equation $(x^3 + xy^2) dx + N(x, y) dy = 0$ is exact.

Solution: The given equation is,

$$(x^3 + xy^2) dx + N(x, y) dy = 0 \rightarrow (1)$$

Here, $M = x^3 + xy^2$

As the given equation is exact $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$$\therefore \frac{\partial}{\partial y} (x^3 + xy^2) = \frac{\partial N}{\partial x}$$

$$\Rightarrow \frac{\partial N}{\partial x} = 2xy$$

$$\Rightarrow N = 2 \cdot \frac{x^2}{2} y + C \quad [\text{Integrating}]$$

$$\therefore N(x, y) = x^2 y$$

Now, $(x^3 + xy^2) dx + x^2 y dy = 0$

$$\Rightarrow x^3 dx + xy^2 dx + x^2 y dy = 0$$

$$\Rightarrow \frac{1}{4} d(x^4) + \frac{1}{2} (2xy^2 dx + 2x^2 y dy) = 0$$

$$\Rightarrow \frac{1}{4} d(x^4) + \frac{1}{2} d(x^2 y^2) = 0$$

$$\therefore \frac{1}{4} x^4 + \frac{1}{2} x^2 y^2 = C. \quad [\text{Integrating}]$$

* Rule of finding the solution of the Exact Differential Equation:

1. If $M(x, y)dx + N(x, y)dy = 0$ is an exact Differential equation, then the solution

$$\int_{y \text{ const}} M(x, y) dx + \int N(x, y) dy = C$$

→ ଯେଉଁଠି x ସଂପର୍କରେ ସମାକଳନ କରାଯାଏ

or, 2. $\int M(x, y) dy + \int_{x \text{ const}} N(x, y) dx = C$

→ ଯେଉଁଠି y ସଂପର୍କରେ ସମାକଳନ କରାଯାଏ

Q. solve the exact differential equation

1. $(y-x)dx + (x+y)dy = 0$

2. $(x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 0$

3. $(\tan y - 3x^2)dx + x \sec^2 y dy = 0$

1. solution: The given equation is

$$(y-x)dx + (x+y)dy = 0 \rightarrow (1)$$

$$\therefore M(x, y) = y-x \text{ and } N(x, y) = x+y$$

$$\Rightarrow \frac{\partial M}{\partial y} = 1, \text{ and } \frac{\partial N}{\partial x} = 1$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ is an exact.}$$

$\therefore$ The solution of equation (1) is

$$\int_{y \text{ const}} (y-x) dx + \int y dy = C$$

$$\Rightarrow xy - \frac{x^2}{2} + \frac{y^2}{2} = C$$

$$\Rightarrow 2xy - x^2 + y^2 = C'$$

Which is the required solution.

2. Solution: The given equation is

$$(x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 0 \longrightarrow (1)$$

$$\therefore M(x, y) = x^3 + 3xy^2 \text{ and } N(x, y) = y^3 + 3x^2y$$

$$\Rightarrow \frac{\partial M}{\partial y} = 6xy \text{ and } \frac{\partial N}{\partial x} = 6xy$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ is an exact.}$$

The solution of equation (1) is

$$\int_{y \text{ const}} (x^3 + 3xy^2)dx + \int y^3 dy = C$$

$$\Rightarrow \frac{x^4}{4} + \frac{3}{2}x^2y^2 + \frac{y^4}{4} = C$$

$$\Rightarrow x^4 + 6x^2y^2 + y^4 = C'$$

Which is the required solution.

3. Solution: The given equation is,

$$(\tan y - 3x^2)dx + x \sec y dy = 0 \longrightarrow (1)$$

$$\therefore M(x, y) = \tan y - 3x^2 \text{ and } N(x, y) = x \sec y$$

$$\Rightarrow \frac{\partial M}{\partial y} = \sec y \text{ and } \frac{\partial N}{\partial x} = \sec y$$

since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so equation (1) is an exact.

The solution of equation (1) is

$$\int_{y \text{ const}} (t \tan y - 3x^2) dx = C$$

$$\Rightarrow x \tan y - x^3 = C$$

Which is the required solution.

Equations Reducible to Homogeneous Form

If $M(x,y) dx + N(x,y) dy = 0 \rightarrow (1)$ is linear but non-homogeneous. Equation (1) in the form,

$$\frac{dy}{dx} = \frac{a_1 x + b_1 y + C_1}{a_2 x + b_2 y + C_2} \rightarrow (2)$$

Case-I: If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = k$ (constant), then the transformation $v = a_1 x + b_1 y$ reduces (2) to a separable equation.

Case-II: If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ then put $x = x + h$, $y = y + k$ to make the equation homogeneous.

Q. solve the non-homogeneous differential equation

1. $(2x - 6y + 3) dx = (x - 3y - 1) dy$

2. $(4x - y + 7) dx = (2x + y - 1) dy$

3. $(6x - 5y + 4) dy = (2x - y + 1) dx$

1. solution: The equation is;

$$(2x - 6y + 3)dx = (x - 3y - 1)dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x - 6y + 3}{x - 3y - 1} \rightarrow (1)$$

$$\text{Here, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{2}{1} = 2$$

$$\text{Let, } v = x - 3y \Rightarrow \frac{dv}{dx} = 1 - 3 \cdot \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{3} \left(1 - \frac{dv}{dx} \right) \rightarrow (2)$$

From (1) & (2), we get

$$\frac{1}{3} \left(1 - \frac{dv}{dx} \right) = \frac{2x - 6y + 3}{x - 3y - 1} = \frac{2v + 3}{v - 1}$$

$$\Rightarrow \frac{dv}{dx} = 1 - \frac{6v + 9}{v - 1} = \frac{-(5v + 10)}{v - 1}$$

$$\Rightarrow \frac{v - 1}{v + 2} dv = -5 dx$$

$$\Rightarrow \left(1 - \frac{3}{v + 2} \right) dv = -5 dx$$

$$\Rightarrow v - 3 \ln(v + 2) = -5x + C$$

$$\Rightarrow x - 3y - 3 \ln(x - 3y + 2) + 5x = C$$

$$\Rightarrow 6x - 3y - 3 \ln(x - 3y + 2) = C$$

$$\Rightarrow 2x - y - \ln(x - 3y + 2) = C'$$

Which is the required solution.

2. Solution: The given equation is

$$(4x - y + 7) dx = (2x + y - 1) dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{4x - y + 7}{2x + y - 1} \quad (1) \quad \left[\text{since } \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \right]$$

$$\text{Let, } x = X + h, y = Y + k$$

$$\Rightarrow dx = dX, dy = dY$$

$$(1) \Rightarrow \frac{dY}{dX} = \frac{4(X+h) - (Y+k) + 7}{2(X+h) + (Y+k) - 1}$$

$$\Rightarrow \frac{dY}{dX} = \frac{4X - Y + 4h - k + 7}{2X + Y + 2h + k - 1} \quad (2)$$

that will be homogeneous if

$$4h - k + 7 = 0$$

$$2h + k - 3 = 0$$

$$\text{here, } h = -1, k = 3$$

$$\therefore x = X - 1, y = Y + 3$$

$$\Rightarrow X = x + 1, Y = y - 3$$

$$\text{From (2)} \Rightarrow \frac{dY}{dX} = \frac{4X - Y}{2X + Y} \quad (3)$$

Which is a homogeneous equation.

$$\text{Let, } Y = VX \Rightarrow \frac{dY}{dX} = V + X \frac{dV}{dX} \quad (4)$$

From (3) & (4), we get

$$V + X \frac{dV}{dX} = \frac{4X - Y}{2X + Y} = \frac{4X - VX}{2X + VX} = \frac{4 - V}{2 + V}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{4-v}{2+v} - v = \frac{4-v-2v-v^2}{2+v} = \frac{4-3v-v^2}{2+v}$$

$$\Rightarrow \frac{v+2}{v^2+3v-4} dv = - \frac{dx}{x}$$

$$\Rightarrow \frac{v+2}{(v-1)(v+4)} dv = - \frac{dx}{x}$$

$$\Rightarrow \left[\frac{1+2}{(v-1)(1+4)} + \frac{-4+2}{(v+4)(-4-1)} \right] dv = - \frac{dx}{x}$$

$$\Rightarrow \frac{3}{5} \cdot \frac{1}{v-1} dv + \frac{2}{5} \frac{1}{v+4} dv = - \frac{dx}{x}$$

$$\Rightarrow \frac{3}{5} \ln(v-1) + \frac{2}{5} \ln(v+4) + \ln x = \frac{1}{5} \ln c \quad [\text{Integrating}]$$

$$\Rightarrow 3 \ln(v-1) + 2 \ln(v+4) + 5 \ln x = \ln c$$

$$\Rightarrow \ln \{ (v-1)^3 (v+4)^2 \cdot x^5 \} = \ln c$$

$$\Rightarrow \left(\frac{y}{x} - 1 \right)^3 \left(\frac{y}{x} + 4 \right)^2 x^5 = c$$

$$\Rightarrow (y-x)^3 (y+4x)^2 = c$$

$$\Rightarrow (y-3-x-1)^3 (y-3+4x+4)^2 = c$$

$$\Rightarrow (y-x-4)^3 (y+4x+1)^2 = c.$$

Which is the required solution. Ans.

3. solution: same as 2.

$$\frac{v_b}{x} = \frac{v_b}{x} \cdot \frac{1}{1} = \frac{v_b}{x} \cdot \frac{1}{\frac{1}{1}} = \frac{v_b}{x} \cdot \frac{1}{\frac{1}{1}} = \frac{v_b}{x} \cdot \frac{1}{1}$$

$$\frac{v_b}{x} = v_b \cdot \frac{1}{x} = v_b \cdot \frac{1}{x} = v_b \cdot \frac{1}{x}$$

$$\frac{v_b}{x} = v_b \cdot \frac{1}{x} = v_b \cdot \frac{1}{x} = v_b \cdot \frac{1}{x}$$

$$\frac{v_b}{x} = v_b \left[\frac{1}{(1-p)(1+v)} + \frac{1}{(1+p)(1-v)} \right] =$$

$$\frac{v_b}{x} = v_b \left[\frac{1}{1-p-v} + \frac{1}{1+p-v} \right] =$$

$$\frac{v_b}{x} = v_b \left[\frac{1}{1-p-v} + \frac{1}{1+p-v} \right] =$$

$$\frac{v_b}{x} = v_b \left[\frac{1}{1-p-v} + \frac{1}{1+p-v} \right] =$$

$$\frac{v_b}{x} = v_b \left[\frac{1}{1-p-v} + \frac{1}{1+p-v} \right] =$$

$$\frac{v_b}{x} = v_b \left[\frac{1}{1-p-v} + \frac{1}{1+p-v} \right] =$$

$$\frac{v_b}{x} = v_b \left[\frac{1}{1-p-v} + \frac{1}{1+p-v} \right] =$$

$$\frac{v_b}{x} = v_b \left[\frac{1}{1-p-v} + \frac{1}{1+p-v} \right] =$$

$$\frac{v_b}{x} = v_b \left[\frac{1}{1-p-v} + \frac{1}{1+p-v} \right] =$$

which is the required solution

Q.E.D.